

1st Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 1201	Geology and Geomorphology of Bengal Basin	2	0	2
ESE 1203	Environmental Earth Sciences	2	0	2
ESE 1202	Environmental Earth Sciences - Lab	0	3	1.5
ESE 1205	Environmental Biology	2	0	2
ESE 1204	Environmental Biology - Lab	0	3	1.5
ESE 1207	Mathematics-II	3	0	3
ESE 1209	Computer Science and IT	2	0	2
ESE 1206	Computer Science and IT - Lab	0	3	1.5
ESE 1211	Sociology for Engineers	2	0	2
ESE 1213	Communicative English	2	0	2
ESE 1208	Field work	0	0	1
ESE 1210	Viva Voce	0	0	1
Total Credits				21.5

2nd Year 1st Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 2101	Hydrology and Hydrogeology	3	0	3
ESE 2103	Oceanography and Limnology	2	0	2
ESE 2105	Ecology	2	0	2
ESE 2102	Ecology- Lab	0	3	1.5
ESE 2107	Environmental Microbiology	2	0	2
ESE 2104	Environmental Microbiology- Lab	0	3	1.5
ESE 2109	Survey and Settlement	2	0	2
ESE 2106	Survey and Settlement-Lab	0	3	1.5
ESE 2111	Soil Mechanics	2	0	2
ESE 2108	Engineering Drawing Lab.	0	3	1.5
ESE 2113	Statistics for Environment	3	0	3
Total Credits				22.0

2nd Year 2nd Semester				
Course Code	Course title	Theory Hrs/Week	Laboratory Hrs/Week	Credit
ESE 2201	Forest and Wildlife Management	3	0	3
ESE 2203	Environmental Geochemistry	2	0	2
ESE 2202	Environmental Geochemistry- Lab	0	3	1.5
ESE 2205	Meteorology and Climatic Process	2	0	2
ESE 2204	Meteorology and Climatic Process –Lab.	0	3	1.5
ESE 2207	Coastal Environment and Blue Economy	3	0	3
ESE 2209	Environmental Biochemistry	2	0	2
ESE 2211	Biodiversity Conservation	2	0	2
ESE 2206	Field Work	0	0	1
ESE 2208	Viva Voce	0	0	1
Total Credits				19.0

ESE 2101. Hydrology and Hydrogeology

Credits: 3; Contact Hours: 3 Hours/Week

Introduction: Definition, Hydrologic Cycle, Importance of Hydrology, Water Balance, Energy Budget, metrological parameters affecting hydrologic cycle, water shed, hydrologic equation.

Precipitation and Water losses: Types and forms, causes and measurements of precipitation and its analysis;

Infiltration, methods determining Infiltration,

Evaporation and transpiration: Evaporation, evaporation as a process, Factors affecting evaporation, methods determining evaporation and evapotranspiration, transpiration, measurement of evaporation, estimation of evaporation.

Run off and flooding: Sources, Catchment characteristics, Factor affecting runoff and runoff estimation Hydrologic Instrumentation, Drainage Patterns; Stream flow Velocity; River-Stage and Discharge; Floods.

Hydrograph: Definition, Hydrograph separation, Unit Hydrograph, Coastal-plain Hydrology.

Storage and groundwater: Origin and occurrence of groundwater, rock properties affecting ground water, subsurface distribution of groundwater, geological formations as aquifers, types of aquifers, Groundwater flow, Darcy's law, Well losses, Well interference, Groundwater recharge, discharge and balance, Methods of Groundwater exploration

Hydrological Environments: Hydrogeological environments of Bangladesh, Sea-water intrusion in coastal aquifers, different types of aquifer in Bangladesh and its problems,.

Mass transport in Aquifers: Advection, Dispersion, Diffusion, Sorption, Retardation, Biodegradation.

Recommended References:

- Todd, D. K., and Mays, L. W. (2005), Groundwater hydrology Ed (Vol. 1625), Wiley, New Jersey. ISBN-13: 978-8126530038.
- Chow, V.T. Handbook of Applied Hydrogeology.
- Gupta, B.L. Water Resources Engineering and Hydrology.
- Kazmann, R.G. Modern Hydrology.
- Meinzer, O.E. Hydrology.
- Shaw, E.M. Hydrology in Practice.
- Todd, D.K. Ground Water Hydrology. Wisler, C.O. and Brater, E.F. Hydrology.

ESE 2103. Oceanography and Limnology

Credits: 2; Contact Hours: 2 Hours/Week

Oceanography

Marine ecosystems: Introduction, features of the oceans; Classification of marine environment (habitat).

Physico-chemical characteristics of marine environment: surface currents, waves, tides, continuity, light, temperature, pressure, viscosity and surface tension, density-temperature relationship, heat capacity, dissolved salts, major and minor ions in sea water, salinity-temperature relationship, dissolved gases and acid/base buffering.

Biota of marine ecosystem and their adaptations: General natures of marine life, Classification of biota of marine ecosystem - marine producers, shore communities, coral reefs - reef forming corals, zonation on coral reefs, symbiotic relationships in coral reef fish, coloration in coral reef fish, classification of marine plankton, migration of plankton, salinity effects on life, temperature effects.

Bay of Bengal: Morphologic features; Bengal Deep-Sea Fan sedimentation; Ecstasy and Relative Sea-Level Changes.

Principles of Palynology and Palynostratigraphy: Classification and distribution of Plants; Palynology of Spores, Pollens, Dinoflagellates etc. and their Paleoecology, Paleoclimatology and Paleoenvironments; Palynology of Bengal Basin and surrounding areas.

Limnology

Introduction: Definition, history and scope of limnology.

Physical features of inland water environments: Density, viscosity, surface film, surface tension, surface radiation, turbidity, effect of light and temperature on the growth of aquatic organisms, thermal stratification and circulation.

Productivity of aquatic habitat: Primary productivity and measurement of primary productivity in aquatic habitats; factors affecting primary production.

Wetlands of Bangladesh (Haors, Baors, Beels and Lakes of Bangladesh) Aquatic environments: classifications, definition of different types of water bodies, freshwater resources of Bangladesh.

Classification of lakes and special type of world lakes; Factors affecting growth and distribution of planktonic organisms.

Recommended References:

- Brasier, M.: Microfossils
- Cushman, A.J. Foraminifera
- Pokorny, V.: Principles of Zoological micropaleontology
- Sumich, J.L. 1992. In Introduction to the Biology of Marine Life, Wm. C. Brown Publishers, 3rd edition.
- Ragothaman, G. 2002. Aquatic Ecology, Trivedy, Agrobios, India.
- Subrahmanyam, N.S. and Sambamurty, A.V.S.S. 2000. Ecology, Narasa Publishing House, NewDelhi, India.
- Steve, N. Oceanography, 1987. John Wiley and Sons.

- Paul, S.W. 1952. Limnology (2nd ed.), McGraw Hill Book Company, USA.
- Ruttner, F. 1975. Fundamentals of Limnology (3rd ed.) University of Toronto Press, Toronto and Buffalo.

ESE 2105. Ecology

Credits: 2; Contact Hours: 2 Hours/Week

Introduction: Concept, definition, branches and scope of ecology;

Ecosystem- Concept and principles of ecosystem; structure, examples and function of ecosystem, Different types of ecosystem; Ecological tools and techniques; Statistical application in ecology;

Energy flow in Ecosystem: Energy in ecosystem, Pattern of flow of energy through the ecosystem and food chains; Biogeochemical Cycles of Ecosystem: Carbon cycle, nitrogen cycle, sulphur cycle and phosphorous cycle; Abiotic and Biotic component of Ecosystem; Phytosociology, Population ecology, Applied ecology.

Ecological factors and their impacts: Introduction, Climatic factors: Light, Temperature, Precipitation, Humidity, Wind-speed and their effects on vegetation; Edaphic factors: Soil types, profiles, components, factors affecting vegetation, soil- water, soil micro-organism and their roles, soil physical and chemical properties; Physiographic factors: Topographic conditions and their effects on vegetation, and Biotic factors: Types, interactions between plants, animals and man, interactions between plants and plants, and interactions between plants and micro-organisms.

Recommended References:

- Odum, E.P. ECOLOGY. Modern Biology Series. Oxford & IBH Publication Co.Ltd. New Delhi.
- Gopal, B. & Bhardwaj, N. ELEMENTS OF ECOLOGY. Vikas Publication Pvt. Ltd. India.
- Shukla, R.S.& Chandel, P.S. 1988. PLANT ECOLOGY AND SOIL SCIENCE. S. Chand & Co. Pvt. Ltd. Ram Nagar, New Delhi, India.
- Verma, P.S.& Agarwal, V.K. 1986. PRINCIPLES OF ECOLOGY.S. Chand & Co. Pvt. Ltd. Ram nagar, New Delhi, India.
- Edward J. Kormondy. CONCEPTS OF ECOLOGY. Prentice-Hall of India Private Limited. New Delhi-110 001.
- Biswarup Mukherjee. Environmental Biology.
- William G. Hopkins. 2002. Introduction to Plant Physiology, 2nd edition, John Wiley and Sons. Inc. New York.

ESE1202. Ecology –Lab

Credits: 1.5; Contact Hours: 3 Hours/Week

Introducing different tools and techniques of population and community study. Identification and preservation of different ecological samples; Field Survey- Determination of productivity of aquatic ecosystem.

Quantification of bloom forming plankton (phytoplankton + zooplankton); Determination of the biomass of producers in the given area; Study the biotic component of a pond ecosystem of JKKNIU Campus.

ESE 2107. Environmental Microbiology

Credits: 2; Contact Hours: 2 Hours/Week

Introduction to Environmental Microbiology: Introduction, Historical perspective, Branches of Environmental Microbiology, Scope of Environmental Microbiology, Modern Environmental Microbiology.

Aquatic Microbiology: Distribution of microorganism in fresh water environments, estuaries and oceans. Domestic and drinking water: pollution, purification, microorganism as indicator of water quality, bacteriological examination of water. Wastewater: characteristics and treatment processes.

Air Microbiology: Microorganisms in air, survival of microorganisms in air, techniques of the study of microorganisms in air, origin of microorganisms in air, control of microorganisms in air.

Soil Microbiology: Porous media, Characteristics of microorganism in porous media, Microbial activities in porous media, Microorganisms in surface soils, Methods of studying soil microorganisms.

Microorganisms in extreme environments: Psychrophiles, Thermophiles, Acidophiles, Alkalophiles, Halophiles and biotechnological applications of Extremophiles and Extremozymes.

Role of microorganisms in biogeochemical cycles: carbon cycle, nitrogen cycle, sulfur cycle and phosphorus cycle. Role of microorganisms in the degradation of hydrocarbons, other organic pollutants and remediation of heavy metals.

Interaction among microorganisms: Mutualism, Commensalism, Antagonism, Competition, Parasitism, Predation.

Recommended References:

- Microbiology: An Introduction. 8th ed. G.J. Tortora, B.R. Funke and C.L. Case. Pearson Education Inc, Singapore, 2007.
- Microbiology. 5th ed. M.J. Pelczar Jr., E.C.S. Chan and N.R. Krieg. Tata McGraw-Hill Publishing Co.Ltd., New Delhi, India,
- Manual of Environmental Microbiology, 2nd ed., C.J. Hurst *et. al.*. ASM Press, Washinton Dc, USA, 2002.
- Environmental Microbiology. R.M. Maier, I.L. Pepper and C.P. Gerba. Academic Press, Elsevier, 2006.
- Microbial Ecology- Fundamentals and Applications, 4th ed., R.M. Atlas and R. Bartha. Benjamin/ Cummings Publishing Co., Inc., 2000.
- Introduction to Soil Microbiology. Martin Alexander. John Wiley & Sons, Inc., 1961
- Soil Microbiology and Biochemistry, E.A. Paul and F.E. Clark, Academic Press, Inc., 1989
- Wastewater Microbiology. Gabriel Bitton. Wiley- Liss, Inc., USA., 1994
- Microbes in Extreme Environments. R. A. Herbert and G. A. Codd, Academic Press, Inc. 1986

ESE 2104. Environmental Microbiology- Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Basic Rules and Requirements of a Microbiology Laboratory; Preparation of different types of culture media; Technique of serial dilution and isolation; Water quality test through faecal coliform counts by streak plate method / pour plate method / spread plate method and enumeration of coliform bacteria; Study of root nodules in legume plants. Sampling of bacteria from air, water and soil; Identification and Culture of microorganisms; Quantitative and qualitative study microorganisms from water, Soil, Fish, Shrimp and other fisheries products and growth study (factors affecting the growth of microorganisms, Temperature, Salinity, Osmotic pressure, pH)

ESE 2109. Survey and Settlement**Credits: 2; Contact Hours: 2 Hours/Week**

Types and Instruments of surveying. **Chain survey:** Plotting details, Errors, Merits and demerits. **Traverse survey:** Procedures, Bearing measure and plotting, Errors and adjustment, Merits and demerits, **Plane Table survey:** Orientation, Merits and demerits, Two and three point problems. **Leveling:** Objectives, Methods, Reduced level, Effects of curvature and refraction and their corrections. **Contouring:** Characteristics and uses, Direct and indirect methods. **Cadastral survey:** instruments, procedures and land settlement and map preparation. **Curves:** Types, Curve components, Setting of circular, transition and vertical curves, Super elevation and Shift. **Land survey:** Land ownership, property rights, laws governing property and definitions of land ownership, mapping and GIS (geographic information systems) software, GPS (global positioning system) handhelds and computer aided design. Analysis of highways and infrastructure. **Project survey:** Classification, Irrigation and Canal projects.

Recommended References:

- Punmia, B. C. (2005), Surveying (Vol. 1), Firewall Media. ISBN-13: 978-81-7008-079-4
- Aziz, M.A. and Shajahan, M. A text book of surveying
- Roy, S. K. (2010), Fundamentals of surveying. 2nd Ed., PHI Learning Pvt. Ltd. ISBN-13: 978-81-203-4198-2

ESE 2106. Survey and Settlement- Lab**Credits: 1.5; Contact Hours: 3 Hours/Week**

Chain survey, Plane table survey, Traverse survey, Leveling and Contouring, Route project, Tacheometry and Stadia surveying, Height and Distance problem, House setting, Curves and Curve settings. Use of Total Station and Global Position Station (GPS).

Recommended References:

- Punmia, B.C. (2005), Surveying (Vol. 1), Firewall Media. ISBN-13: 978-81-7008-079-4
- Aziz, M.A. and Shajahan, M. A text book of surveying
- Roy, S.K. (2010), Fundamentals of surveying. 2nd Ed., PHI Learning Pvt. Ltd. ISBN-13: 978-81-203-4198-2

ESE 2111: Soil Mechanics**Credits: 2; Contact Hours: 2 Hours/Week**

Definition of soil, volume composition of soil: Soil as a component of ecosystem; Minerals of soils, Importance of soil. Soil as a disperse three phase system, Mass volume relationship of soil constituents.

Soil structure and Morphology: Mechanism of aggregate formation and dispersion; classification on the basis of shape; importance of soil structure in relation of air-water content; Soil densities and pore space.

Stress and strain processes: Stress-strain relations and failure of soil bodies, Soil strength and its measurement, Soil compression, Soil compaction in relation to wetness, shear stress, stability of slopes in soils.

Soil water; Concepts of soil water, Factors affecting soil water retention, Methods of measuring soil water-gravimetric, tensiometer and neutron scattering method, Soil water potential-metric potential, pressure potential, gravitational potential, osmotic potential, Measurement of metric potential by tensiometer method

Water, contaminant and gas movement in soil: Infiltration-definition, factors affecting, equations, methods of measurement, Hydraulic conductivity-definition, Darcy's law, methods of measurement, classes of hydraulic conductivity, contaminant mass flux, movement of exchanges of gases in soil, solute transport in soil.

Soil temperature: Importance, factor affecting and management of soil temperature, thermal properties of soils, heat flow equation.

Mechanical analysis: Principle and methods, Derivation of Stoke's law.

Soil consistency and tillage: Forms of soil consistency, swelling, shrinkage, Tillage and puddling, Plough pan-formation, effects on soils and crops.

Groundwater hydrology: Groundwater flow, vertical flow of water, pollutant flow through groundwater, diffuse and point pollution, groundwater drainage and pollution, flow of confined and unconfined aquifers,

Recommended References:

- Daniel Hillel, 1998. Environmental Soil Physics
- Keller, E.A., 1992, Environmental Geology 6th ed.
- Graig, R. F., 2004: Graig's Soil Mechanics. London
- Punmia, B. C., 1994: Soil Mechanics and Foundations. Laxmi Publications Pvt. Ltd., New Delhi.
- Harrison, J. P and Hudson, J. A.: Engineering Rock Mechanics Part II

- Skinner, B.J. & Porter, S.C., 1995, The Dynamic Earth, 3rd edition; John Wiley & Sons, Inc. New York.
- Wild, A., 1996, Soils and the Environment, Cambridge Univ. Press.
- Miller, R.W. & Donahue, R.L., 1997, Soils in our Environment, Prentice Hall Private Limited.
- The Nature and Properties of Soils. Brady and Weil, 13th edition.
- Fundamentals of Soil Science, H.D. Foth, 6th edition, John Wiley and Sons.
- Fundamentals of Soil Physics, D. Hillel, 1st edition Academic Press, USA.

ESE 2108: Engineering Drawing-Lab

Credits: 2.0; Contact Hours: 3 Hours/Week

Fundamental Concepts: Drawing instruments and their uses; Lettering, numbering and constructing title strip, adopting symbols; Views; Projections: First angle, Third angle; Generation of views of solid bodies in different planes, Sectional views, Auxiliary views, Isometric views, Dimensioning, Basic concept of working drawing.

AutoCAD: Importance to design and drafting, Setting up a drawing: starting AutoCAD, menu, planning for a drawing, basic commands, making a simple 2-D drawing, layers, object snap, poly lines and other features, file handling and display control, editing and dimensioning.

Recommended References:

- I H Morris: Geometrical Drawing
- Hemanta Kumar Bhattacharia: Pratham Engineering Drawing
- R.W.P Aroogaswamy: Mining Geology

ESE 2113. Statistics for Environment

Credits: 3; Contact Hours: 3 Hours/Week

Frequency distribution: Measures of central tendency, dispersion; moments, skewness and Kurtosis. **Correlation and Regression:** Bivariate data, correlation and regression coefficients, regression. Probability: Sample space event, probability of events, theorem, of total and compound probability, conditional probability, bayes theorem. **Random variable:** Probability function, distribution function, joint, marginal and conditional probability function, expectation and moment generating function. **Parent Distributions:** Binomial, poisson, negative binomial, normal and exponential distribution. **Sampling Distributions:** χ^2 , t and F distributions. **Test of Hypothesis:** Test for population, mean, variance, correlation coefficient and regression coefficients.

Recommended References:

- Spiegel M R, Theory and Problems of Statistics
- Hoel P G, Elementary Statistics
- Hogg R V & Craig A T, Introduction to Mathematical Statistics, 4th Ed, Macmillan, London
- Mood, Graybill & Boes, Introduction to the Theory of Statistics, 3rd Ed, McGraw Hill, NY
- Mostafa M G, Methods of Statistics
- Milnyk M, Principles of Applied Statistics
- Barlow R G, Statistics
- Loveday, Practical Statistics and Probability
- Mosteller, Rourke & Thomas, Probability with Statistical Applications, 2nd Ed, Addison-Wesley, USA
- Goon A M & Gupta M N, Fundamentals of Statistics, Vol-I